



Md. Omar Sha Rafi

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RESEARCH INTEREST

My research focuses on natural products chemistry and phytochemistry, with particular emphasis on the isolation, characterization, and bioactivity evaluation of plant-derived bioactive compounds. I am interested in exploring multidisciplinary approaches that integrate phytochemical screening, analytical characterization, molecular docking, and in-vitro/in-vivo biological assays to identify and validate novel therapeutic agents from traditional medicinal plants. My work aims to bridge traditional herbal knowledge with modern drug discovery and pharmacological science, contributing to the development of biologically active molecules.

RESEARCH EXPERIENCE

- **Supervision:** Trained 6 MSc and 12 undergraduate students in compound isolation, organic synthesis, and analytical techniques.
- **M.S. Thesis:** Isolation of Phytoconstituents, Fatty Acid Analysis and Biological Investigation on the Flower of *Acmella calva* (DC.) R.K. Jansen (Surjo Konna)
 - Applied **spectroscopic and chromatographic methods** (NMR, UV-Vis, IR, XRD, HPLC) for structural elucidation and biological activity studies.
- **B.S. Project:** Phytochemical Screening, In-Vitro Cytotoxicity Evaluation and Fatty acid analysis of *Spilanthes acmella* L. Flower

PUBLICATIONS

Publications.

- Mimi, S. S., Fahim, M. S. Z., Rafi, M. O. S., Mia, M. S., & Hasan, M. K. (2026). Experimental and computational evaluation of bioactive secondary metabolites from *Trewia nudiflora* stem bark. *Fitoterapia*, 195, 107479.
- Al Mamun, A., Shanchary, H. R., Rafi, M. O. S., Ray, S. K., Shandhi, S. P., Bhuiyan, R. H., Chowdhury, F., & Muslim, T. (2026). Innovative TiO₂-SiO₂-La₂O₃ nanocomposite systems for enhanced wastewater treatment performance. *Next Nanotechnology*, 10, 100807.
- Meem, M. I., Fahim, M. S. Z., Mia, M. S., *Rafi, M. O. S., & Akhter, K. (2026). Catechol substitution governs antioxidant potency in 5-arylidene-thiazolidine-2, 4-diones: Synthesis, structure-activity relationships, DFT, docking, and molecular dynamics. *Results in Chemistry*, 103793.
- Al Mamun, A., Ray, S. K., *Rafi, M. O. S., Shandhi, S. P., Bhuiyan, R. H., Chowdhury, F., & Muslim, T. (2026). Design, characterization, and application of a mixed metal oxide (Al₂O₃-La₂O₃-V₂O₅) nanocomposite in photocatalytic degradation of methylene blue: a multi-technique approach. *RSC advances*. (Scopus Indexed)
- Ahmed, M. S., *Rafi, M. O. S., & Muslim, T. (2026). Valorization of crab shell waste into chitin, chitosan, and carboxymethyl chitosan: Extraction, chemical modification, and characterization. *Next Materials*, 11, 101710. (Scopus Indexed)
- Fahim, M. S. Z., Moulick, S. P., Mimi, S. S., Dey, S. S., Rafi, M. O. S., Chhanda, S. A., ... & Muslim, T. (2026). Exploring the bioactivity of *Ageratum conyzoides* L.: Antioxidant, Antimicrobial, and Cytotoxic Properties of Flower Extracts. *Next Research*, 102216.
- *Rafi, M. O. S., Shristy, N. T., Mimi, S. S., Chhanda, S. A., Rahman, M. H., Rahman, M. S., ... & Muslim, T. (2025). Antioxidant, thrombolytic, cytotoxic and membrane stabilizing activities and fatty acid composition of *Acmella Calva* (DC.) Flower. *Bangladesh Journal of Botany*, 54(3), 433-439. (Scopus Indexed)

- Hossain, R. B., Mimi, S. S., *Rafi, M. O. S., Chhanda, S. A., Ibrahim, M., & Muslim, T. (2026). In vivo Evaluation of Analgesic, Antidiarrheal, Hypoglycemic, and Anti-Inflammatory Activities of *Sarcolobus carinatus* Methanolic Extract in Swiss Albino Mice. *Bangladesh Pharmaceutical Journal*, 29(2), 282-289.
- Bristy, R. R., Mimi, S. S., Chhanda, S. A., Ali, M. H., Rafi, M. O. S., & Muslim, T. (2025). Investigation of the Pharmacological Potential of *Staurogyne zeylanica* Kuntze: In vitro and In vivo Studies. *Bangladesh Pharmaceutical Journal*, 28(1), 28-36.
- Jahan, I., Ali, M. H., Shristy, N. T., Rafi, M. O. S., Mimi, S. S., Siddik, M. N. A., ... & Chowdhury, T. A. (2024). Assessing Therapeutic Potentials of *Catharanthus roseus* (L.) G. Don Focusing on Anti-Diabetic, Analgesic and Anti-Diarrheal Activities. *Bangladesh Pharmaceutical Journal*, 27(2), 215-222.
- Shristy, N. T., Rafi, M. O. S., Mimi, S. S., Chowdhury, V., Rahman, M. H., & Chowdhury, T. A. (2024). In vitro Biological Activities and Fatty Acid Composition Analysis of *Musa paradisiaca* L. Inflorescence. *Dhaka University Journal of Pharmaceutical Sciences*, 23(1), 69-76.
- Rafi, M. O. S., Shristy, N. T., Chowdhury, V., Rahman, M. H., Mimi, S. S., Afroz, R., & Chowdhury, T. A. (2023). *Acmella calva* (DC.) R.K Jansen flower: A comprehensive in vivo study of its central and peripheral analgesic, hypoglycemic, antidepressant, and anti-diarrheal properties. *Dhaka University Journal of Pharmaceutical Sciences*, 22(1), 65-71.

Conference Presentations

- Rafi, Md. Omar Sha; Rahman, Md. Hasanur; Chowdhury, Tofail Ahmad; “Isolation of Phytoconstituents and Biological Investigation on the Flower of *Acmella calva* (DC.) R.K. Jansen (Surjo Konna)” *BCSIR Congress on Integrated Approach for Adapting 4IR* December 1-3, 2022, Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhaka, Bangladesh (Oral Presentation)
- Shristy, Nusrat Tabassum; Rafi, Md. Omar Sha; Chowdhury, Vaskar; Rahman, Md. Hasanur; Chowdhury, Tofail Ahmad; “Phytochemical and pharmacological investigations of the inflorescences of *Musa paradisiaca* L. (Atia kola)” *BCSIR Congress on Integrated Approach for Adapting 4IR* December 1-3, 2022, Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhaka, Bangladesh (Oral Presentation).

WORK EXPERIENCE

Lecturer of Chemistry

Department of Chemistry, University of Dhaka

Dec 2023–Present

- **Led 3 externally funded research projects (PI/Co-I), securing ~1 million BDT**, producing 3 publications.
- **Supervised 6 MSc and 12 undergraduate students** in compound isolation, organic synthesis, and advanced analytical techniques.
- **Authored/co-authored 12 peer-reviewed publications.**
- **Designed and taught 2+ theory courses and 4 lab courses** in Organic and General Chemistry, enhancing student learning outcomes.

Research Fellow

Bose Center for Advanced Study and Research in Natural Sciences, University of Dhaka

01/07/2023 – 01/12/2023

Project Title: Evaluation of Phytochemicals and Bioactive Properties of *Adensoma indianum*.

Research Assistant

Centennial Research Grant (CRG 2nd Phase), University of Dhaka

01/07/2022 – 30/06/2023

Project Title: Investigation of Nutritional Content, Phytochemical and Biological Activity on the Fruit of *Limonia acidissima* L.

EDUCATION

MS in Organic Chemistry, University of Dhaka | CGPA 3.91/4.00

Aug 2021 - Dec 2022

BS in Chemistry, University of Dhaka | CGPA 3.81/4.00

Jan 2016 – Mar 2021

SKILLS

- Software: MS-Word, MS-Excel, MS PowerPoint, SigmaPlot, ChemDraw, PyRx, Discovery Studio, Autodock Vina
- Instrumental: FT-IR, UV Spectrophotometer, GC-FID
- Language: Bangla, English
- Programming Skill: Python 3, Javascript, HTML, Ruby, Bash
- Others: Excellent presentation and communication skills, Good teaching skills, Team work, Critical thinking, Good organization and management skills, Analytical and problem-solving skills, Synack red team member

RESEARCH AND GRANTS

- **Principal Investigator:** *Isolation of Secondary Metabolites, Fatty Acid Analysis and Biological Investigation on the Flower of Acmella calva (DC.) R.K. Jansen (Surjo Konna)*, Funded by Ministry of Science and Technology (MOST), Bangladesh, 80,000 BDT. Experimental study exploring therapeutic potential.

Outcomes: Resulted in 1 peer-reviewed publication and the training of 2 MSc students.

- **Principal Investigator:** *Phytochemical and Pharmacological Exploration of Trianthema portulacastrum L. Leaves: A Potential Medicinal Weed*, Funded by Ministry of Science and Technology (MOST), Bangladesh, 80,000 BDT. Experimental study exploring therapeutic potential.
- **Co-Investigator:** *Synthesis and Structure-Activity Relationships of Thiazolidinedione Derivatives as Potential Therapeutic Agents*—Funded by University Grants Commission (UGC), Bangladesh, 750,000 BDT. Contributed to project design, experimentation, and data analysis.

HONORS & AWARDS

- National Science and Technology (NST) fellowship (December, 2021)
- University Merit Scholarship Based on academic excellence at B. S (April, 2022)
- Mitsubishi Corporation International Scholarship (2018)
- Dhaka Board Government Merit Scholarship award based on academic excellence at Secondary School Certificate (2014)
- Dhaka Board Government Merit Scholarship award based on academic excellence at Junior School Certificate (2010)
- Dhaka Board Government Merit Scholarship award based on academic excellence at Primary Scholarship exam (2007)
- 1st runner up at HackerOne BUG HUNT 2023 organised by bug bounty community Bangladesh at ICT Division, Agargaon, Dhaka

Courses: Takes 3 courses and conducted 4 different academic labs

CH 485 – Medicinal Chemistry (Optional Major, 4th Year)

- **Credits:** 2
- **Focus:** Introduces the principles of drug design, structure–activity relationships, and biochemical mechanisms of therapeutic agents.
- **Key Topics:**
 - Pharmacodynamics and pharmacokinetics
 - Synthetic strategies for drug molecules
 - Case studies of antibiotics, anticancer agents, and cardiovascular drugs
- **Outcome:** Students gain insight into how chemistry underpins modern medicine and pharmaceutical innovation.

CMG 100 – Fundamentals of Chemistry (Minor, 1st Year)

- **Credits:** 4

- **Focus:** A foundation course for non-major students covering the basics of chemical principles.
- **Key Topics:**
 - Atomic structure and bonding
 - States of matter and chemical reactions
 - Introduction to acids, bases, and solutions
- **Outcome:** Provides essential chemical literacy for students in other science disciplines.

CMGL 100 – General Chemistry Laboratory (Minor, 1st Year)

- **Credits:** 2
- **Focus:** Practical training in basic chemical experiments for non-major students.
- **Key Topics:**
 - Qualitative and quantitative analysis
 - Simple synthesis and purification techniques
 - Laboratory safety and data recording
- **Outcome:** Builds hands-on skills and reinforces theoretical knowledge from CMG 100.

CHL 324 – Organic Chemistry Laboratory II (Major, 3rd Year)

- **Credits:** 4
- **Focus:** Advanced laboratory work in organic chemistry.
- **Key Topics:**
 - Multi-step organic synthesis
 - Isolation and characterization of natural products
 - Use of spectroscopic techniques (IR, NMR) for structural analysis
- **Outcome:** Students develop competence in modern organic lab methods and analytical techniques.

CHL 463 – Industrial Chemistry Laboratory (Major, 4th Year)

- **Credits:** 2
- **Focus:** Practical exposure to industrial chemical processes.
- **Key Topics:**
 - Pilot-scale synthesis and process optimization
 - Environmental and safety considerations in industry
 - Analytical methods for industrial products
- **Outcome:** Prepares students for careers in chemical industries by linking academic knowledge with real-world applications.

□ **Carbohydrates and Glycoconjugates: ORG 526 Description:** Part of the M.S. Organic Chemistry curriculum, this course provides advanced knowledge on the chemistry, synthesis, and biological roles of carbohydrates and their conjugates. While the specific detailed syllabus page was not fully displayed in the snippets, it is categorized as a 3-credit advanced course within the Organic Chemistry thesis group.

Teaching Philosophy

To me, teaching is not just about explaining a subject or completing a syllabus. It is about helping students grow in confidence, think independently, and develop a genuine interest in learning. A classroom should be a place where students feel comfortable asking questions, sharing ideas, and learning from mistakes without fear of judgment.

My approach to teaching focuses on students as active participants in the learning process. I try to make lessons clear, engaging, and meaningful by connecting academic concepts to everyday life and real situations. In chemistry, many topics can feel abstract at first, so I place strong emphasis on building understanding step by step. Discussions, examples, practical work, and problem-solving activities help students see how theory connects with the world around them.

I also understand that every student learns differently. Some grasp ideas quickly through explanation, while others need visuals, practice, or one-on-one guidance. Because of this, I try to use a variety of teaching methods and create an inclusive environment where all students feel supported and capable of succeeding. I value respect, collaboration, and open communication in the classroom.

Assessment, in my view, should do more than assign grades. It should help students recognize their progress, understand where they need improvement, and build confidence in their abilities. Through

quizzes, assignments, presentations, and feedback, I encourage students to take responsibility for their own learning.

Education should also prepare students for life beyond the classroom. Along with academic knowledge, I hope to help students develop integrity, resilience, communication skills, and a sense of responsibility toward others. These qualities are just as important as technical expertise.

As a teacher, I believe learning never stops. I continue to reflect on my teaching, improve my methods, and stay informed about new developments in both education and my field. This helps me grow professionally and better support my students.

At the heart of my teaching philosophy is a simple belief: every student can learn and improve when given the right guidance, encouragement, and opportunity.

Supervised Projects:

2025:

Name: Nirupama Das Tumpa

Exam Roll No: 2428506

Project Title : Optimization of Cassava Starch/PVA Bioplastic Films via Glutaraldehyde Cross-linking and a Comparative Study of Mechanical and Physical Properties

MD SHAON MIA

Examination Roll No.-2428575

Class Roll: SH-053

Project Title : Design, Synthesis, Characterization and Computational Insights of Some Modified Dihydropyrimidine Derivatives

Noor-E Alam Raihan

Roll: 2428561

Project Title : Green Synthesis, Characterization, and Clinical Relevance of CuO and PbO Nanoparticles: Antimicrobial, Antioxidant, Cytotoxicity, Anti-inflammatory, Thrombolytic Assay

Name: Ahsan Ullah Miraj

Examination Roll: 2428566

Project Title: Synthesis, Isolation and Characterization of a Benzimidazole derivative

Name: Zerin Tasnim Rahman summa

Roll: RK-087

Title: Bioactive potential of Persicaria flaccida with multi- Analytical approach

Name: Md Safkat Zaman Fahim

Roll: AE-032 Title: Evaluation of phytochemical and bioactive properties of ageratum conyzoides flower extracts

Thesis Supervision:

Name: Abdullah Al Mamun

Thesis title: Design and Characterization of Metal Oxide Catalysts for the Efficient Production of Furfural from Sugarcane Bagasse-Derived Hemicellulose

Name: Imrul Hasan Bhuiyan

Thesis title: Green synthesis and Characterization of CuO Nanoparticles Using Mesua ferrea

Leaf Extract and Evaluation of Biological and Photocatalytic Activities

Name: Alamgir Hossain Session

Thesis title :study of phytochemical investigation and biological activity evaluation of *Dentella repens*

Name: Md Ariful islam

Thesis title :Comprehensive Study on Isolation and Characterization of Bioactive Compounds, Fatty Acid Profiling, Phytochemical Constituents, and Biological Activities of *Sargassum muticum*”

Name: Rifka Hossain

Thesis title: Study on biological activity, compound isolation and characterisation and fatty acid analysis of *Sarcolobus carinatus*.

Name: Md. Safkat Zaman Fahim

Thesis title: Green Synthesis and Comparative Physicochemical and Biological Evaluation of Iron Oxide (Fe_2O_3) and Cobalt Ferrite (CoFe_2O_4) Nanoparticles Using *Neptunia oleracea* Extract

Name: Zarin Tasnim Rahman Summa

Thesis title: Phytochemical Investigation of plant *Persicaria flaccida*: Isolation and Structural Characterization of Secondary Metabolites and Evaluation of In Vitro and In Vivo Biological Activities

Name: Md. Rakib Hasan

Thesis title: Synthesis and Computational Validation of Aminated Chitosan for Enhanced Dye Adsorption and Biological Transport

Name: Khaironnesa Baly

Thesis title: Bioactivity-Oriented Phytochemical Investigation of *Careya arborea* Roxb. Flowers

Name: Zakia Nazmin

Thesis title: Bioactivity-Oriented Phytochemical Investigation of *Careya arborea* Roxb. Flowers